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WILD SOUTH AMERICAN CAMELIDS (MAMMALIA, ARTIODACTYLA): A REVIEW OF THEIR EVOLUTIONARY AND BIOGEOGRAPHIC HISTORY

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ABSTRACT—This review examines the evolutionary and biogeographic history of South American camelids, which include the wild species guanaco (*Lama guanicoe*) and vicuña (*Vicugna vicugna*), and their domestic counterparts, llama (*L. glama*) and alpaca (*V. pacos*). We trace their origins within the Camelidae family, which appeared in North America approximately 46 million years ago. This family experienced at least four major radiation events during the Eocene and Miocene, culminating in the diversification of the subfamily Camelinae into the tribes Camelini and Lamini. The ancestors of South American camelids emerged from the Lamini tribe, dispersing into South America and surviving the widespread extinctions that affected many Camelidae during the Late Pleistocene and Early Holocene. This paper reevaluates the taxonomic classification of the Lamini using both paleontological and molecular evidence, with a specific focus on the lineages that led to the modern South American species. We also explore how environmental changes—particularly climatic shifts following the Last Glacial Maximum and the expansion of modern humans—have shaped the biomes inhabited by these unique camelids. This analysis provides a comprehensive overview of the key evolutionary milestones that have defined the unique history of this iconic group.

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INTRODUCTION

Camelids (Camelidae; Gray, 1821) are a diverse family of mammals within the order Artiodactyla (Owen, 1848). Their evolutionary history began in the Holarctic region, with the earliest known fossils dating to the Eocene epoch, approximately 46 million (Ma) years before present (BP) in North America (Honey et al., 1998). A series of diversification events from the Eocene and the Pliocene led to the two extant tribes: Camelini and Lamini (Harris et al., 2010; Honey et al., 1998; Rybczynski et al., 2013; Scherer, 2013). The Camelini tribe (Gray, 1821) includes the dromedary (*C. dromedarius*) and the Bactrian camel (*C. bactrianus*), iconic desert species of Africa and Asia, respectively, alongside the wild Bactrian camel (*C. bactrianus ferus*) of Central Asia (Burger, 2016). The Lamini tribe (Webb, 1965), often referred to as South American camelids (SACs), comprises two wild species—the guanaco (*Lama guanicoe*) and vicuña (*Vicugna vicugna*)—and two domesticated species—the llama (*L. glama*) and alpaca (*V. pacos*) (Díaz-Maroto et al., 2021; Fan et al., 2020). The guanaco is the largest SAC (90–120 kg) with the widest geographic range, spanning from

northern Peru (8°S) down the Andes to Tierra del Fuego and Navarino Island (55°S), and from sea level to 4000 m altitude (Franklin, 1982; González et al., 2006; Iranzo et al., 2022; Lucherini et al., 2000). In contrast, the smaller vicuña (38–46 kg) is exclusively found in the high-altitude Puna grasslands (3800–4900 mbsl) in the Central Andes (9°50'S–29°15'S) (Franklin, 1982; Marín et al., 2007). Genetic evidence confirms that the domestication of these species began 6000 to 7000 years ago, with the llama arising from the guanaco and the alpaca from the vicuña (Wheeler, 1995; Yacobaccio, 2021).

Camelids are a unique group of ungulates that possess distinct anatomical, physiological, and biochemical adaptations which contribute to their resilience in harsh and resource-limited environments. They are classified in the suborder Tylopoda (Illiger, 1811) due to their unique foot structure: instead of hard hooves, they feature two toes with hardened, flexible soles. Additionally, their digestive system differs markedly from that of true ruminants (e.g., cattle, sheep, deer). While both are foregut fermenters, a camelid's foregut is partitioned into three compartments (C1, C2, and C3), which notably all contain glandular tissue and lack the characteristic papillae of a ruminant stomach (Lechner-Doll et al., 1995; Vallenás et al., 1971). These structural, motility, and biochemical differences are linked to greater efficiency in absorbing solutes and water from the foregut compared with ruminants (Rocha et al., 2021; Rutagwenda et al., 1989). The ability to recycle nitrogen from

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urea when dietary protein is scarce represents a key adaptation that allows camelids to survive in nutrient-poor environments (Lechner-Doll et al., 1995). Comparative genomic studies reveal that the Camelini and Lamini tribes share conserved adaptations suited to desert environments that include resilience to heat stress, and resistance to ultraviolet radiation (Bas & Bonacic, 2003; Wu et al., 2014).

The unique morphological and physiological traits of camelids have enabled them to thrive in challenging environments, including mountain ecosystems and arid or semi-arid regions (González et al., 2006; León et al., 2024; Marín et al., 2007; Wu et al., 2014; Xue et al., 2021). This exceptional success in adapting to harsh biomes likely facilitated their early domestication by human communities in these territories (Díaz-Moroto et al., 2021; Wu et al., 2014; Yaccobaccio, 2021). Today, camelids continue to be vital for the livelihoods and cultural heritage of many Indigenous Peoples and Local Communities (IPLCs) residing in demanding environments, such as the Gobi and Atacama Deserts (Burger, 2016; Faye, 2020; Vilá & Arzamendia, 2022; Wu et al., 2014). In South America specifically, SACs are particularly important to Andean cultural identity, offering essential support for both food security and economic opportunities for the local development of communities (FAO, 2024; Vilá & Arzamendia, 2022).

Beyond their social importance, camelids are megaherbivores with crucial ecological roles. Their ecosystem functions are multifaceted, encompassing the dispersion of native plant seeds, the regulation of invasive species populations, and the maintenance of trophic structures within food webs (Pringle et al., 2023). For instance, successful rewilding initiatives in Chile have leveraged guanacos to aid the ecological restoration of degraded areas (Root-Bernstein et al., 2017, 2024). However, despite their resilience, camelids face several threats that make them vulnerable. Like other large mammals in extreme climates, they are impacted by ongoing climate change and habitat loss driven by the expansion of the livestock industry (Flores et al., 2023; Hetem et al., 2014). Furthermore, they are affected by legal hunting and trade in regions such as Argentine Patagonia (Barbuto, 2025), as well as the risks associated with the scientifically unsupported relocation of individuals for restoration projects (Balza et al., 2023; Guerisoli & Ovejero, 2024).

This review synthesizes key findings from paleontology, ecology, and phylogenetics to examine the evolutionary and biogeographic history of wild South American camelids (SACs). We discuss how major climatic environmental shifts influenced their primary biomes and conclude by highlighting critical knowledge gaps and suggesting future research directions to advance understanding of SAC biogeography and autecology.

ORIGIN AND MIGRATORY EVENTS OF SUBFAMILY CAMELINAE

Poebrotherium wilsoni Leidy, 1847 is consistently regarded as one of the earliest recognized members of Camelidae, ranging from the late Eocene to the early Oligocene (see Fig. 1A). It is frequently utilized as ancestral stock in phylogenetic analyses of the family (Honey et al., 1998; Scherer, 2013). Nevertheless, discoveries of older, scarcer fossils such as the Uintan *Protylopus* (Wortman, 1898) and *Poebrodon* (Gazin, 1955) have obscured the direct ancestry of *Poebrotherium*. In their systematic revision, Honey et al. (1998) consider *Poebrodon* as the most primitive camelid known.

Camelid evolution is marked by four major radiation events (Honey et al., 1998). The origin of the modern subfamily, Camelinae (Gray, 1821), occurred during the fourth radiation, in the Early Miocene, approximately 17.5–14 Ma (Fig. 1B). As

Camelinae diversified, other older camelid subfamilies gradually disappeared, becoming extinct by the Late Miocene (Honey et al., 1998). Camelinae subsequently diverged into two tribes: Camelini (Old-World camelids) and Lamini (New-World camelids) (Carmanchahi et al., 2022; Harrison, 1985; Honey et al., 1998). While the fossil record initially suggested this divergence occurred between 16 and 12 Ma BP (Honey et al., 1998), the exact timing remains a subject of ongoing debate. More recent fossil and molecular evidence support a divergence event between 17.5 and 16 Ma BP (Heintzman et al., 2015; Lynch et al., 2020). However, some mitochondrial DNA studies suggest an earlier split dating the divergence to around 25 Ma BP, during the Arikarean North American land mammal age (Cui et al., 2007). Following this divergence, both the tribes Camelini and Lamini gave rise to various genera across North and South America (Burger, 2016; Scherer, 2013).

The subfamily Camelinae was the only camelid group documented to migrate from North America to Eurasia, Africa, and South America (Harris et al., 2010; Honey et al., 1998). Members of the Camelini dispersed into Eurasia via the Bering Strait land bridge during the Late Miocene (Made van der & Morales, 1999) (Fig. 1C). *Paracamelus* is the best documented example of this successful dispersal, crossing the Bering Strait multiple times during the Pleistocene and surviving in the Arctic zone until approximately 7000 years ago (Rybczynski et al., 2013). Based on fossils recovered across Asia, Europe, and Africa, *Paracamelus* is considered the most likely ancestor of *Camelus* (Buckley et al., 2019; Harris et al., 2010; Rybczynski et al., 2013). Additionally, other Camelini taxa, such as *Camelops*, occupied eastern Beringia during Pleistocene interglacial periods when forests and shrublands were present (Zazula et al., 2017).

Other research supports Webb's (1974) hypothesis that some Lamini taxa migrated to South America via the Isthmus of Panama around 2.7 Ma years ago during the Great American Biotic Interchange (GABI) (Labarca et al., 2013; Webb, 1991; Wheeler, 2012; Woodburne et al., 2006) (Fig. 1E). In North and Central America, at least five species assigned to the genus *Hemiauchenia* are recorded from the Late Miocene through the Late Pleistocene (Cisneros, 2005; Laurito & Valerio, 2016; Scherer, 2013). The oldest South American fossil of *Hemiauchenia* dates to the Chapadmalalan South American Land mammal age (SALMA), approximately 3.55–2.59 Ma BP (Gasparini et al., 2017). The genus *Palaeolama* is likewise reported from both North and Central America during the Late Pliocene–Early Pleistocene (Bravo-Cuevas & Jiménez-Hidalgo, 2015; Cisneros, 2005; Pérez, 2013). It has been hypothesized that northern *Palaeolama* species migrated southward during isolated GABI events, whereas southern forms may have dispersed northward (Gasparini et al., 2017; Scherer, 2013; Scherer et al., 2007) (Fig. 1F).

Five genera—*Eulamaops*, *Hemiauchenia*, *Palaeolama*, *Lama*, and *Vicugna*—are recognized as having diversified across the Pampas and Andean regions during the Late Pliocene and Early Pleistocene (Gasparini et al., 2017; Scherer, 2013) (Fig. 1F). Genomic analyses indicate that *Lama* and *Vicugna*, the only extant genera, diverged from a common ancestor between 2 and 3 Ma BP (Fan et al., 2020; Wheeler, 1995) (Fig. 1G). More than 30 Lamini species have been proposed based on fossils from the Pliocene and Pleistocene. The taxonomic classification proposed by Honey et al. (1998) was later re-evaluated by Scherer (2009) for South American taxa, who synonymized a significant number of previously described forms within this clade (Labarca et al., 2013). Currently, the classification proposed by Scherer (2009, 2013) is widely accepted for the tribe Lamini. All genera and species (northern and southern) described for this tribe are summarized below and detailed further in the following brief section on paleontological systematics.

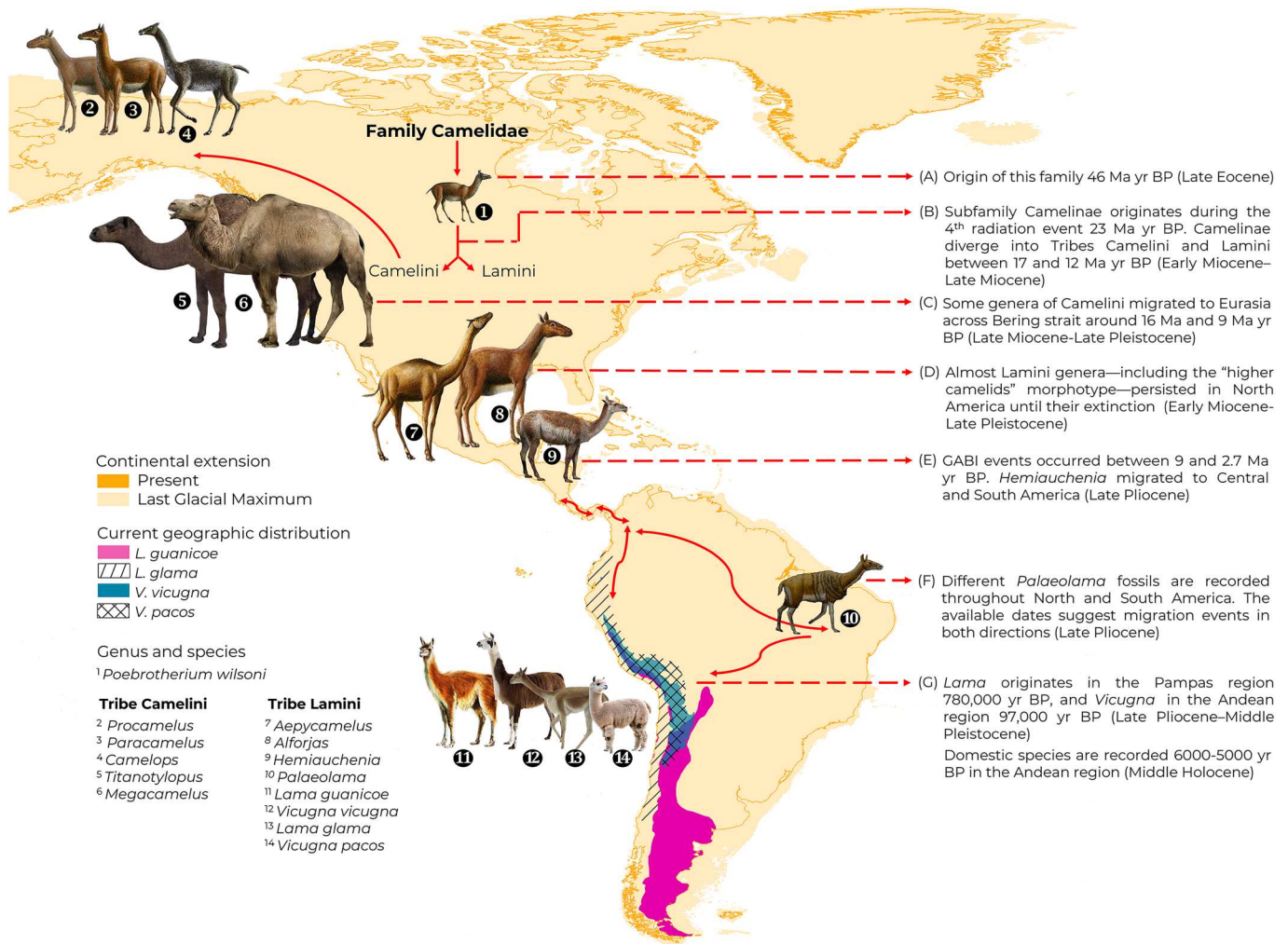


FIGURE 1. Summary of the main events in the evolutionary and biogeographic history of the subfamily Camelinae. Each genus is represented by a reconstruction of its appearance based on morphological features and approximate body size. The distribution ranges of *Lama guanicoe* and *Vicugna vicugna* were mapped using IUCN polygons, whereas the ranges of *Lama glama* and *Vicugna pacos* were drawn following McLean & Niehaus (2022).

SYSTEMATIC PALEONTOLOGY

Order ARTIODACTYLA Owen, 1848
 Suborder TYLOPODA Illiger, 1811
 Family CAMELIDAE Gray, 1821
 Subfamily CAMELINAE Gray, 1821
 Tribe LAMINI Webb, 1965

AEPYCAMELUS Macdonald, 1956(†)

Location and Temporal Range—North America, Late Miocene to Late Pleistocene. *Aepycamelus* is attributed to the “higher camelid” morphotype, distinguished by the elongation of the vertebrae and limbs and the development of certain molar structures (e.g., the metastylid) (Honey et al., 1998). A phylogeny reconstructed by Lynch et al. (2020) positioned this genus as a successor to *Pleiolama*, though the study concludes that further evidence is needed to confirm this phylogenetic relationship. Fossil species include *Aepycamelus alexandrae* Davidson, 1923, *A. elrodi* Douglass, 1909, and *A. robustus* Leidy, 1858.

ALFORJAS Harrison, 1979 (†)

Location and Temporal Range—North America, Middle Miocene to Late Pleistocene. This genus belongs to the “higher camelids” morphotype. Recognized species include *Alforjas magnifontis* Gregory, 1942, and *A. taylora* Harrison, 1979.

BLANCOAMELUS Dalquest, 1975 (†)

Location and Temporal Range—North America, Pliocene to Pleistocene. This genus was mainly distributed in central and southern North America (Janis et al., 1998). Fossil records of *Blancocamelus* are sparse, with most findings originating from the Blacan locality (Honey et al., 1998). The recognized taxon is *Blancocamelus meadei* Dalquest, 1975.

EULAMAOPS Ameghino, 1889 (†)

Location and Temporal Range—South America, Late Pleistocene. This genus has been documented in the Pampas region of Argentina during Lujanense, a SALMA dated between 126,000 and 11,000 years BP (Scherer, 2009, 2013). The unique recognized species is *Eulamaops parallelus* Ameghino, 1884.

HEMIAUCHENIA Gervais & Ameghino, 1880 (†)

Location and Temporal Range—North America, Late Miocene to Early Holocene. This genus migrated to South America via the Isthmus of Panama around 2.7 Ma years ago (Labarca et al., 2013). It has been suggested that the ancestral form of *Hemiauchenia* gave rise to *Hemiauchenia paradoxa* in South America during the Late Pliocene (Scherer, 2009; Webb & Stehli, 1995). Based on the fossil record *Hemiauchenia* is further regarded as the ancestral lineage from which *Palaeolama*, *Lama*, and *Vicugna* later emerged (Scherer, 2013; Wheeler, 2012). Recognized species include *H. blancoensis* Meade, 1945, *H. gracilis* Meachen, 2005, and *H. edensis* Frick, 1921, in North America, although the last two are not well known and require revision (Scherer, 2013). Additionally, *H. macrocephala* Cope, 1893, and *H. vera* Matthew, 1909, are recorded from North to Central America (Cisneros, 2005; Laurito & Valerio, 2016), while in South America *H. paradoxa* Gervais & Ameghino, 1880, is recognized, along with the recent addition of *H. mirim* Greco et al., 2022, described from Brazil's intertropical region.

LAMA Frisch, 1775

Location and Temporal Range—South America, Late Pliocene to Present. Fossil records indicate that *Lama* has been present in Argentina (Late Pliocene to Early Holocene), Bolivia (Middle Pleistocene), Chile, Brazil, and Uruguay (Late Pleistocene) (Scherer, 2009; Scherer et al., 2007). Modern populations are found in Chile, Argentina, Peru, Bolivia, and Paraguay (González et al., 2006). Recognized species are *Lama guanicoe* Müller, 1776, and *L. castelnaudi* Gervais, 1855, originating in the Andean region and later dispersing southward (Scherer, 2013). Other species that have been described for this genus include *L. owenii* (Gervais & Ameghino, 1880), *L. angustimaxila* (Ameghino, 1889), and *L. gracilis* (Gervais & Ameghino, 1880), all assigned to the Late Pleistocene of southern Patagonia (Nami & Menegaz, 1991; Villavicencio et al., 2016). However, the first two species are currently grouped within the same large-bodied morphotype of *L. cf. owenii* (Nami & Menegaz, 1991), while Scherer's (2009) analysis indicates that *L. angustimaxila* and *L. owenii* are synonymous with *H. paradoxa*. *L. gracilis*, in turn, has been reassigned to *Vicugna vicugna* (Labarca, 2020; Villavicencio et al., 2016; Weinstock et al., 2009). Domesticated species include *L. glama* Linnaeus, 1758, which was derived from *L. guanicoe* (Díaz-Maroto et al., 2021; Fan et al., 2020; Marín et al., 2017).

PALAEOLAMA Gervais, 1867 (†)

Location and Temporal Range—North America, Late Pliocene to Early Holocene. Scherer (2013) suggests that this genus evolved in North America during the Late Pliocene (~2 Ma years ago) and migrated to South America during the GABI events, following the earlier migration of *Hemiauchenia* (Gasparini et al., 2017). This genus persisted in South America from the Late Pliocene to the Late Pleistocene, becoming extinct by the Early Holocene (Gasparini et al., 2017). Several species have been described, including *Palaeolama mirifica* Simpson, 1929, in North America, and *P. hoffstetteri* Guérin & Faure, 1999, *P. major* Liais, 1872, and *P. weddelli* Gervais, 1855, in South America. Other fossil records attributed to *P. aequatorialis* (Hoffstetter, 1952) and *P. niedae* (Guérin & Faure, 1999) are considered synonymous with *P. major*; just like *P. reissi* (Branco, 1883) and *P. crassa* (Hoffstetter, 1952), both are synonymous with *P. weddelli* (Scherer, 2009).

PLEIOLAMA Webb & Meachen, 2004 (†)

Location and Temporal Range—North America, Late Miocene to Early Holocene. *Pleirolama* is regarded as the earliest Lamini genus (Scherer, 2013). Its oldest occurrence is

approximately 11 Ma years ago (Harrison, 1985; Meachen, 2005). Previously, it was also referred to as *Pliauchenia* (Wheeler, 1995) before this name was invalidated by Webb & Meachen (2004). Fossil species assigned to this genus are *Pleirolama mckennai* Webb & Meachen, 2004 and *P. vera* Webb & Meachen, 2004. Although Wheeler (1995) proposed that both *Hemiauchenia* and *Alforjas* evolved from *Pleirolama*, recent analyses indicate the genus is polyphyletic, with *P. vera* positioned as an early member of the Lamini (Lynch et al., 2020).

VICUGNA Gray, 1872

Location and Temporal Range—South America, Middle Pleistocene to Present. Fossil records of *Vicugna* span Argentina, Bolivia, Chile, southern Brazil, and Uruguay from the Middle to Late Pleistocene (Cartajena et al., 2010; Labarca & Prieto, 2009; Scherer (2013); Weinstock et al., 2009). Recognized species include *Vicugna vicugna* Molina, 1782, *V. provicugna* Boule & Thevenin, 1920, and *V. pacos* Linnaeus, 1758, which was domesticated from *V. vicugna* (Fan et al., 2020; Marín et al., 2017). *Vicugna vicugna* and *V. pacos* are extant taxa with populations in Chile, Argentina, Peru, and Bolivia (Marín et al., 2007). Recent analyses have reassigned fossils previously attributed to *L. gracilis* (Gervais & Ameghino, 1880) to *V. vicugna* based on morphological, metric, and genetic evidence (Labarca & Prieto, 2009; Scherer, 2009; Weinstock et al., 2009).

DISCUSSION

Pleistocene–Holocene Transition: Environmental Change and Extinction Events

Not all Lamini and Camelini taxa successfully navigated the environmental and climatic shifts spanning the Late Pleistocene and Early Holocene (Labarca, 2020) (Fig 1D). This transitional period included significant events such as the Last Glacial Maximum (LGM, approximately 26,000–19,000 years BP) and the arrival of modern humans in the Americas (Del Papa et al., 2024; Gruhn, 2020; Martini et al., 2017). These factors are thought to have contributed to the severe impacts of the Late Quaternary Extinction (LQE) on mammalian diversity across both North and South America (Barnosky et al., 2004, 2016; Koch & Barnosky, 2006; Stuart, 2015).

In North America, for instance, the Arctic population of *Camelops* went extinct during the early Wisconsin glaciation (around 75,000 years BP) (Zazula et al., 2011). Remaining populations, restricted to temperate regions, became extinct around 13,000 years BP, notably after facing hunting pressure from early humans (Buckley et al., 2019; Zazula et al., 2017). Overhunting by modern humans is considered the primary driver of this extinction (Heintzman et al., 2015).

Similar pressures also affected members of the tribe Lamini, as several North and South American species disappeared from the fossil record during the Late Pleistocene, except for the extant genera *Lama* and *Vicugna*. Major environmental changes have been documented in three key biomes where significant Lamini fossil records have been found: the Pampas ecosystem (30–39°S); Andean region represented by the Puna, Sechura, and Atacama Deserts and ecosystems associated with the South American Arid Diagonal (SAAD), between 5–25°S; and Patagonia (38–56°S) (Morrone, 2015). The paleontological and archeological evidence for each region and the main environmental changes are briefly summarized below (see Fig. 2).

Pampas Region

The fossil record from Argentina's Pampas plains indicates the presence of *Hemiauchenia* between the Late Pliocene and Late

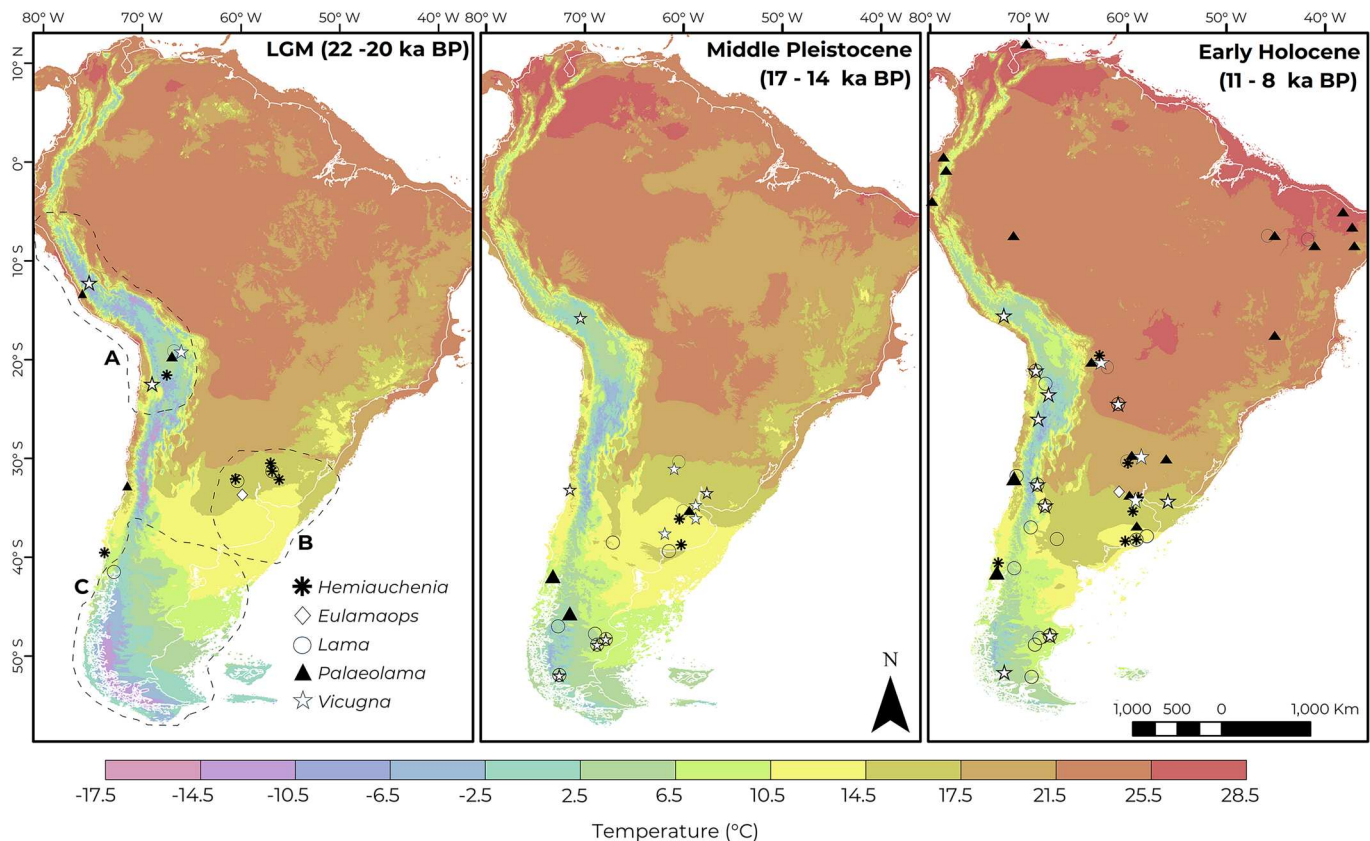


FIGURE 2. Representation of climatic variation and the occurrence of South American camelids across three key biomes analyzed in this study (dash polygons): **A**, Andes; **B**, Pampas; and **C**, Patagonia. Mean annual temperature changes (color scale) are shown for three periods: the Last Glacial Maximum, the Middle Pleistocene (ACR and CAPE events), and the Early Holocene (onset of human settlements), using climate layers from the CHELSA project (Karger et al., 2023). Fossil occurrences correspond to paleontological and archeological evidence compiled from public repositories, including the Paleobiology Database (PBDB, paleobiodb.org) and the New and Old World Fossil Mammal Database (NOW, nowdatabase.org), as well as from the published literature (Scherer, 2013).

Pleistocene (Scherer, 2013), with evidence suggesting its role in the origin of *Lama* in this area during the Middle Marplatán (SALMA) dated to approximately 3–1.2 Ma BP (Gasparini et al., 2017). Records also point to a possible derivation of *V. vicugna* from *L. guanicoe*, first appearing in the Ensenadense (SALMA) dated between 1.2–0.8 Ma BP (Menegaz et al., 1989; Scherer, 2013). The region further preserves Late Pleistocene remains of *Vicugna*, *Palaeolama*, and *Eulamaops* (de Almeida Rocha-dos et al., 2017; Gasparini et al., 2017; Scherer, 2009, 2013).

In this region, paleoclimate studies reconstruct shifts from the LGM to the Holocene based on palynological and sedimentological evidence from paleolakes (Iriarte, 2006; Prieto, 2000). Records show colder LGM conditions linked to glacial advances ~23,000 years ago (Cuña-Rodríguez et al., 2020), a drier phase during the Antarctic Cold Reversal (ACR) 14,700–13,000 years ago (Pedro et al., 2016), and humid intervals associated with the Central Andes Pluvial Event (CAPE) between 12,800–11,000 years BP (Cuña-Rodríguez et al., 2020). These changes drove vegetation shifts from C3 grass-dominated steppes in the Late Pleistocene to C4 vegetation in the Early Holocene (Iriarte, 2006; Lopes et al., 2023; Prieto, 2000).

Human presence is documented in this region since at least 12,000 years BP, exploiting large mammals like guanacos (Gutiérrez & Martínez, 2008), though zooarchaeological data show earlier occupations during the LGM (Del Papa et al., 2024).

Andean Region

Paleontological evidence indicates that *Palaeolama* inhabited this region from the Late Pliocene to Early Pleistocene in Peru (Salas et al., 2003) and from the Early to the Middle Pleistocene in Bolivia (Gasparini et al., 2017). A questionable Late Pliocene–Early Pleistocene record of *Hemiauchenia* has been reported from Jujuy (Scherer, 2013). Fossils of *L. guanicoe* suggest its presence around the Altiplano since at least 780,000 years ago (Cajal et al., 2010), with its distribution likely shaped by repeated glacial advances and retreats until the LGM ~25,000 years BP (Carmanchahi et al., 2022). The oldest records of *V. vicugna* are from Tarija, Bolivia, dated to ~97,000–73,000 years BP (Wheeler & Laker, 2009).

Paleoclimate reconstructions reveal intercalary humid phases in hyperarid environments, such as the Atacama Desert, which fostered paleolake formation between the Late Pliocene and Middle Pleistocene (Evenstar et al., 2017; Ritter et al., 2018). Between 18–22°S, conditions during the LGM (14,000–12,000 years BP) included stable temperatures and double the modern rainfall (CAPE events), favoring large paleolakes and richer vegetation cover (Ammann et al., 2001; Kull et al., 2002). Such resource availability, combined with the extreme aridity of the SAAD, likely drove *V. vicugna* to occupy higher elevations north of this barrier (Wheeler & Laker, 2009).

Human occupation of the Peruvian Andes is documented from ~9000 years BP (Aldenderfer, 1999), with evidence of specialized hunting of vicuñas and guanacos from ~6000 years BP (González et al., 2019). Early domestic forms appeared around 6000–5500 years BP, followed by the development of pastoral societies centered on alpacas and llamas from ~5500 years BP (Díaz-Maroto et al., 2021; Wheeler, 1999).

Patagonia

In Chile, camelid records coincide with the LGM ~17,000 cal years BP (Labarca, 2020; Metcalf et al., 2016), with the oldest Lamini evidence being a *L. guanicoe* footprint dated between 52,300 and 44,300 years BP in the north of Patagonia (Campos-Medina et al., 2022). In Argentina, *L. guanicoe* dates to ~12,800 years BP in Santa Cruz (Miotti et al., 2003) and ~11,000 years BP in Tierra del Fuego (González et al., 2006). *V. vicugna* appears in archeological sites from 13,000–10,000 years BP in Magallanes and Tierra del Fuego (Villavicencio et al., 2016; Weinstock et al., 2009; Wheeler & Lake, 2009), while *Hemiauchenia* is documented at Pilauco, its southernmost record (Labarca 2020; Labarca et al., 2013).

Climate reconstruction indicates that megafauna colonized southern Chilean Patagonia following the LGM amid repeated glacial retreats and advances linked to the ARC ~14,800–12,800 years BP (Barnosky et al., 2016; Sagredo et al., 2011). During this period, the landscape was dominated by tundra vegetation, consisting of grasses and herbs adapted to cold conditions. This climatic instability is hypothesized to have contributed to the extinction of *L. gracilis* (reassigned to *V. vicugna*) around 15,000 years BP (Villavicencio et al., 2016). Subsequently, the open grasslands were replaced by *Nothofagus* forests around 11,000 years BP (Barnosky et al., 2016; Moreno et al., 2012; Villavicencio et al., 2016). This major ecological transition likely promoted the decline of ungulates strongly associated with grasslands, such as *Hippidion saldiasi* and a large Lamini identified as *L. cf. owenii* (Gervais & Ameghino, 1880) (Villavicencio et al., 2016; Villavicencio et al., 2019). Critically, this extinction period coincided with the establishment of human communities in the region (15,000–14,000 years BP), whose hunting pressure on large mammals and increased fire frequency further contributed to population declines (Barnosky et al., 2016; Moscardi et al., 2020; Pérez et al., 2016). Despite these several environmental and anthropogenic pressures, the guanaco became the only camelid to persist in Patagonia, even while constituting a crucial component of the human diet during the Holocene (Metcalf et al., 2016; Rindel et al., 2021; Villavicencio et al., 2016).

In summary, when examining the evidence across all regions, the ‘Combination Theory,’ understood as the synergistic effect of climate change and human overhunting, appears to plausibly explain the pattern of megafauna extinctions in both North and South America (Barnosky et al., 2004; Koch & Barnosky, 2006; Stuart, 2015). Consequently, by the Early Holocene, nearly all South American Lamini genera were extinct, with the exceptions of *Lama* and *Vicugna* (Gasparini et al., 2017; Labarca, 2020; Scherer, 2013).

The Modern South American Camelids (SACs)

As previously noted, the guanaco and vicuña are the only wild camelids that survived in South America following the major climatic and environmental changes of the Pleistocene–Holocene transition (Scherer, 2013). Their distinct biogeographic histories are partly reflected in the fossil record. The fossil distribution of the guanaco largely mirrors its extensive modern range (González et al., 2006; Marín et al., 2013; Politis et al., 2011; Rindel et al., 2021). In contrast, vicuña fossils are found predominantly within their current high Andean distribution, although Pleistocene

records also exist further south, including the Magallanes region (Cajal et al., 2010; Labarca, 2020). Importantly, morphological, metric, and ancient DNA studies have led to the reassignment of many fossils previously identified as *Lama gracilis* (Gervais & Ameghino, 1880) to the vicuña lineage (Labarca, 2020; Labarca & Prieto, 2009; Scherer, 2009; Weinstock et al., 2009). Recent discoveries attributed to *L. gracilis* in Patagonia (Villavicencio et al., 2016), Tierra del Fuego (Weinstock et al., 2009), central Chile (Quinteros Commune) (López-Mendoza et al., 2018), and northwestern Argentina (Tucumán) (Torres-Carro et al., 2021) significantly expand the known Pleistocene distribution of vicuñas.

Guanacos and vicuñas have coexisted sympatrically without evidence of significant antagonistic interspecific interactions throughout their shared range (Cajal, 1989; Lucherini et al., 2000; Villavicencio et al., 2016). This expanding and complex fossil record, however, raises several key questions regarding their late Quaternary history: (i) why did only guanacos persist and thrive in Patagonia?; (ii) why did other contemporary Lamini taxa disappear entirely?; (iii) how widespread was the vicuña distribution during the Pleistocene?; and (iv) when and why did vicuñas become restricted primarily to high Andean habitats? (Weinstock et al., 2009).

While climatic shifts partly explain these divergent outcomes, ecological strategies also played a decisive role in shaping the survival and distribution of both species. Vicuñas disappeared from the Patagonian fossil record around 15,000 years BP due to unfavorable conditions induced by the ACR (Villavicencio et al., 2016). Conversely, the Puna of the Central Andes might have served as a climatic refuge, where vicuñas established populations during the Late Pleistocene (Wheeler & Lake, 2009). This region offered stable temperatures and diverse vegetation resulting from wetter periods between ca. 14,000 and 12,000 years BP (Wheeler & Lake, 2009). However, the South American Arid Diagonal—the diagonal strip of arid land running from the coastal Pacific desert in current Peru to the southern Atlantic Patagonian coast in Argentina—presents a significant biogeographic barrier in this region (González et al., 2019). Survival in the extreme high-elevation environments occupied by vicuñas today requires specific physiological adaptations, including enhanced hemoglobin-oxygen affinity, small red blood cells, high capillary density, and a greater heart-to-body weight ratio compared with other camelids (Jürgens et al., 1988; Monge & León-Velarde, 1991; Wheeler & Lake, 2009; Yamaguchi et al., 1987). On the other hand, by analyzing the autecology of these species, guanacos are generalist herbivores capable of thriving in diverse habitats and consuming a wide variety of vegetation (González et al., 2006). Conversely, vicuñas are highly selective feeders specializing in microthermal grasses characteristic of arid zones, rather than species exclusively adapted to high mountain life per se (Menegaz et al., 1989). The post-LGM shift in the distribution of various grass groups drove the geographic retraction and eventual extinction of vicuña populations in Patagonia (Menegaz et al., 1989), while having minimal impact on the widely distributed guanaco (Cajal et al., 2010). Additionally, guanacos exhibit greater flexibility in adapting to environmental changes, including undertaking seasonal migrations and forming larger mixed groups when conditions require it (González et al., 2006). In contrast, vicuñas display marked territoriality and a strong preference for specific grassland habitats (González et al., 2019). These contrasting life-history strategies may have been crucial during the environmental fluctuations of the Pleistocene–Holocene transition, potentially leading to vicuña niche restriction (Weinstock et al., 2009).

These ecological and evolutionary traits must be examined carefully considering the fragmented distributions and distinct genetic lineages of each species (González et al., 2006; Marín et al., 2013).

Molecular studies have shown that different geographic populations present clear genetic differences: *L. g. cacsilensis* distributed north of the SAAD (8–44°S); *L. g. guanicoe* south of the SAAD, mainly in Patagonia (18–55°S); and a third hybrid population distributed in a contact zone between the first two (26–32°S) (Carmanchahi et al., 2022; González et al., 2013); and *V. v. mensalis* that inhabits the humid Puna (9°50'–18°30'S); *V. v. vicugna* in the dry Puna (21°30'–29°15'S); and a third hybrid population distributed in the contact zone between the first two (18°30'–21°30'S; González et al., 2019, 2020). Additionally, recent analyses of population distribution and connectivity patterns indicate that climatic and anthropogenic pressures will intensify soon unless effective short-term conservation measures are implemented (Castillo et al., 2018; Espinosa et al., 2018; Mata et al., 2024; Mesas et al., 2023).

Final Remarks and New Research Opportunities

Despite the rich paleontological and archeological record of the Camelidae across the Americas, the taxonomy of tribes Camelini and Lamini, requires continuous revision (Burger, 2016; Greco et al., 2022; Laurito & Valerio, 2016). There are particularly acute challenges within the Lamini tribe, where over 30 Pliocene–Pleistocene species have been proposed, often based on South American fossils with uncertain diagnoses (Scherer, 2013). Recent studies strikingly illustrate this taxonomic instability. For instance, a key fossil previously assigned to *Palaeolama* was recently reclassified as *Hemiauchenia mirim*, a newly identified species from the intertropical region of Brazil (Greco et al., 2022). Similarly, the reassignment of *Lama gracilis* fossils to the vicuña lineage has significantly expanded the inferred Pleistocene distribution of vicuñas beyond their current high-altitude range (Torres-Carro et al., 2021; Weinstock et al., 2009). Each reclassification directly affects migration hypotheses across the entire Lamini tribe, necessitating reevaluation of the proposed biogeographic patterns for South America. The constant influx of new fossils and molecular datasets, including genome-scale analyses, necessitates the ongoing reevaluation of the biogeographic history of the New World camelids (Gasparini et al., 2017; Lynch et al., 2020; Scherer, 2013).

In this context, our review provides a comprehensive synthesis of paleontological and ecological evidence for the Lamini, integrating disparate lines of research to clarify long-term patterns in diversity, distribution, and extinction. By also summarizing the key environmental events across South America, this work contributes to reconstructing the biogeographic history of camelids and understanding how the geographic features, particularly the Andes Mountain range, have shaped the distribution of each Lamini species (González et al., 2016; Marín et al., 2013; Wheeler & Laker, 2009). Integrating major climatic chronologies across the SACs range allows us to contextualize the severe Pleistocene–Holocene transition pressures—such as the LGM, the ACR, and the arrival of humans—that contributed to the extinction patterns observed among Lamini in the Neotropics. The severity of this period is further underscored by the disappearance of other American megafauna including major ungulate families such as Equidae and Gomphotheriidae (Labarca, 2020; Lima-Ribeiro et al., 2013; Villavicencio et al., 2019). Despite this widespread extinction event, several critical questions remain regarding the survival and current distribution of the two wild South American camelids. First, what factors allowed the guanaco and vicuña to be the sole surviving Lamini species? Second, if both species potentially had broader and overlapping distribution ranges during the Pleistocene, why is the vicuña now largely restricted to the high-altitude Puna of the central Andes? Finally, how has the guanaco managed to maintain its wide distribution range despite the profound climatic and environmental changes of the Holocene?

Answering these questions requires analytical approaches that integrate evolutionary and ecological perspectives, as well as their interactions. Paleoecological studies often rely on stable isotope analysis and dental wear, when samples and resources allow it. Such methods have provided valuable insights into the paleodiets of South American ungulates, contributing to a better understanding of their ecological niches and the vegetation-climate dynamics of different time periods (de Oliveira et al., 2021; Domingo et al., 2012; Martínez-Polanco et al., 2024). To date, however, no comparable analyses have been conducted in SACs.

Ancient DNA and genomic studies offer another powerful line of evidence. In camelids, these approaches have shed light on the evolutionary history of the family and clarified key processes such as domestication (Díaz-Maroto et al., 2021; González et al., 2019; Fan et al., 2020; León et al., 2024; Marín et al., 2007, 2013, 2017).

Ecoinformatic tools, particularly Environmental Niche Models (ENMs) and Species Distribution Models (SDMs), also provide robust frameworks for evaluating spatial and temporal changes in species distributions under major environmental shifts such as the LGM (Fordham et al., 2020; Svenning et al., 2011). These models are increasingly supported by open-access databases, such as the Paleobiology Database (PBDB, paleobiodb.org) and the New and Old World Database of Fossil Mammals (NOW, now-database.org), which compile extensive fossil records along with well-resolved taxonomic and geochronological information (McClennen et al., 2017; Žliobaitė et al., 2023). In parallel, several initiatives have generated paleoclimatic datasets and simulations at long time scales, further enriching the possibilities for integrative analyses (Brown et al., 2018; Fordham et al., 2017; Karger et al., 2023). Over the past decades, the use of ecological modeling to explore species' responses to past climate change has gained momentum, particularly in the context of defaunation events such as the LQE in the Americas (Araújo et al., 2021; DeSantis et al., 2012; Hadly et al., 2009; Martínez-Meyer et al., 2004; Villavicencio et al., 2019). However, in this era of interdisciplinary research and technological advances, combining paleoecological, genetic, and modeling approaches offers an unprecedented opportunity to obtain more robust results and explore broader temporal and spatial scales.

Ultimately, understanding past changes in species distributions is essential to reduce uncertainty when predicting the impacts of accelerated climate change on remaining biodiversity (Fordham et al., 2020). Such knowledge can also guide more effective management actions, including the identification of climate refugia, ecosystem restoration (Root-Bernstein et al., 2017; Root-Bernstein & Guerrero-Gatica, 2024), and the translocation of individuals or populations to regions they once inhabited (Guerisoli & Ovejero, 2024), thereby strengthening the prospects for long-term conservation.

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AUTHOR CONTRIBUTIONS

AGC conducted the bibliographic search, created the figures, and drafted the manuscript. All authors edited the manuscript.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are openly available in The New and Old Worlds Dataset (NOW) at URL: https://nowdatabase.luomus.fi/species_list.php and The Paleobiology Database (PDB) at URL: http://paleobiodb.org/data1.2/occs/list.csv?datainfo&rowcount&base_name=Camelidae&taxon_reso=species&ident=all&pgm=gplates,scotese,seton&show=attr,cl ass,coords,loc,paleoloc,geo.

DISCLOSURE STATEMENT

No potential conflict of interest was reported by the author(s).

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